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ONE BOTTLE BATTERY

SOLID-STATE SODIUM-ION BATTERIES FROM UPCYCLED PET PLASTIC WASTE

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THE PROBLEM



Imported Li-Ion Dependency

India's battery landscape is dominated by lithium-based chemistries constrained by scarce domestic reserves of lithium and cobalt, leading to high costs, geopolitical risk, and supply chain vulnerability.



3.4 Million Tonne PET Crisis

India generates 3.4 million tonnes of plastic waste annually. PET recycling often results in downcycling to low-value materials — a massive untapped resource sitting idle.



Defence Readiness Gap

Imported lithium-ion batteries pose fire risks at frontlines. Standard batteries fail in extreme cold conditions of Siachen and Ladakh, creating a critical tactical vulnerability.



Technological Vassalage

Continued dependence on foreign battery technology forces India to import billions in energy storage annually, creating a permanent strategic vulnerability in the nation's clean energy transition.

THE SOLUTION

Transforming India's plastic waste crisis into a sovereign energy asset — one PET bottle at a time.

01

COLLECT

Waste PET bottles sourced from India's 3.4M tonne plastic stream — our strategic mineral reserve

02

UPCYCLE

Chemical solvolysis converts PET into high-purity Disodium Terephthalate (DST) salts and hard carbon anodes

03

POWER

Assembled into solid-state sodium-ion batteries — non-flammable, ballistic-safe, and 100x cost-disruptive

WHY SODIUM-ION? THE STRATEGIC ADVANTAGE

100X

COST DISRUPTION
vs. Lithium-Ion

Metric	Lithium-Ion (LIB)	Sodium-Ion (SIB)	India's Gain
Raw Material	Rare / Imported	Abundant / Domestic	Energy Sovereignty
Precursor Cost	\$14L–\$80L / tonne	\$20K–\$25K / tonne	Cost-Effective EVs
Current Collector	Copper (Anode)	Aluminium (Both)	Reduced System Weight
Safety Profile	High Thermal Risk	Enhanced Stability	Tropical Climate Ready
Supply Chain	Highly Concentrated	Diversified / Local	Resilience to Shocks

SIB technology reduces cell-level costs by an estimated 20–30% relative to LFP chemistries, with a path to 100X disruption through PET upcycling.

~1000x

More Abundant
than Lithium

20–30%

Cost Reduction
vs. LFP

500 GW

India's 2030
Renewable Target

THE BREAKTHROUGH: TOTAL MOLECULAR UPCYCLING



SINGLE-SOURCE BATTERY ENGINEERING

Every battery component — anode, electrolyte, salts — derived from one PET bottle. True circular manufacturing.



SAFETY-FIRST CHEMISTRY

Non-flammable solid electrolytes. Zero thermal runaway. Ballistic-safe — engineered for defence and tropical India.



COLD-START RESILIENCE

Operates from -70°C to +45°C. Maintains 92% capacity at -40°C — critical for Siachen & Ladakh frontlines.



100X COST DISRUPTION

PET plastic feedstock costs near-zero vs. \$14L–80L/tonne for lithium carbonate. Giga-scale game changer.

THE "ONE BOTTLE" BATTERY

3-PHASE DEVELOPMENT ROADMAP

PHASE 1

COMPLETED ✓

Polymer Electrolyte Dev

- ▶ Alkaline battery prototype via PET solvolysis
- ▶ Disodium terephthalate (DST) synthesis protocol
- ▶ Rechargeable PoC demonstrating reversible Na-ion intercalation
- ▶ Patent applied for PET → liquid electrolyte conversion
- ▶ Theoretical research for solid state sodium ion completed

PHASE 2

FUNDING REQUIRED

Solid-State R&D

- ▶ Year 1: Advanced anode engineering & feedstock optimization
- ▶ Year 2: Solid-state electrolyte synthesis (NASICON + PET matrix)
- ▶ Year 3: Full PoC battery assembly & performance validation
- ▶ Target: 130–250 Wh/kg, 500+ cycles, -20°C to 60°C range

PHASE 3

COMMERCIALIZATION

Pilot & Scale (Post R&D)

- ▶ Shift from lab-scale to industrial multi-layer pouch cells (2–5 Ah)
- ▶ Strategic Pilot Plant and Self-Reliant R&D Laboratory
- ▶ Integration with PLI scheme & National Critical Mineral Mission
- ▶ ARAI/ICAT certification for Indian EV market deployment
- ▶ Giga-scale manufacturing roadmap for defence & EV sectors

PHASE 2 R&D PLAN: YEAR-BY-YEAR

YEAR 1 <i>ANODE ENGINEERING</i>	YEAR 2 <i>SOLID ELECTROLYTE</i>	YEAR 3 <i>PoC FABRICATION</i>
Q1 PET collection & solvolysis optimization → Standardized carbon precursor batches	Q5 NASICON filler synthesis via sol-gel → High-purity ceramic electrolyte powders	Q9 Full-cell assembly → 4–5 working solid-state sodium batteries
Q2 Multi-stage pyrolysis (800–1600°C) → Optimized hard carbon powders	Q6 PET-derived polymer electrolyte films → Self-standing flexible membranes	Q10 Safety & abuse testing (Thermal, Overcharge) → Safety compliance report
Q3 Characterization (XRD, SEM, Raman, XPS, BET) → Structural & porosity data reports	Q7 Conductivity & ESW optimization → Electrolytes with target ionic conductivity	Q11 Long-term cycling & post-mortem analysis → Degradation mechanism insights
Q4 Electrochemical half-cell benchmarking → Capacity & rate capability data	Q8 Interface engineering & compliance tests → Stable electrode-electrolyte protocols	Q12 PoC demonstration & final project reporting → Final technical & financial report

MARKET OPPORTUNITY & TARGET CUSTOMERS

18.1%

CAGR

Global SIB EV Market
2026–2034

1.3 TWh

DEMAND

India Battery Storage
Needed by 2047

₹3,400 Cr+

OPPORTUNITY

Indigenous Industry
Replacing Imports



Electric 2W & 3W Manufacturers

Ola Electric, Ather Energy, Hero MotoCorp — seeking low-cost LFP alternatives for price-sensitive Indian market



Renewable Energy Storage (BESS)

Tata Power, Reliance New Energy — large-scale stationary storage for solar/wind grid management



Telecom Infrastructure

Backup power for towers in remote/high-temperature zones where traditional batteries fail or require expensive cooling



Defence & Rural Micro-Grids

Frontline military electronics in extreme cold + rural solar homes under Atmanirbhar Bharat initiative

ACHIEVEMENTS & TEAM



Alkaline Battery Prototype

First working PET-derived alkaline cell — proof the upcycling concept is viable



Rechargeable PoC Built

Demonstrated reversible Na-ion intercalation & electrochemical stability across multiple cycles



Patent Filed

IP secured for PET plastic → liquid electrolyte conversion mechanism



Theoretical research

► Theoretical research for solid state sodium ion completed



TN Government Recognition

Featured & interviewed by top regional channel at state-level startup event



EDII-TN recognition

Recognized under the Innovation Voucher Program by EDII Tamil Nadu

THE FOUNDING TEAM



Ms. Princy Polin S

Co-Founder & Chief Executive Officer

Leads product vision, technology strategy, and investor relations. Spearheading India's indigenous sodium battery revolution.



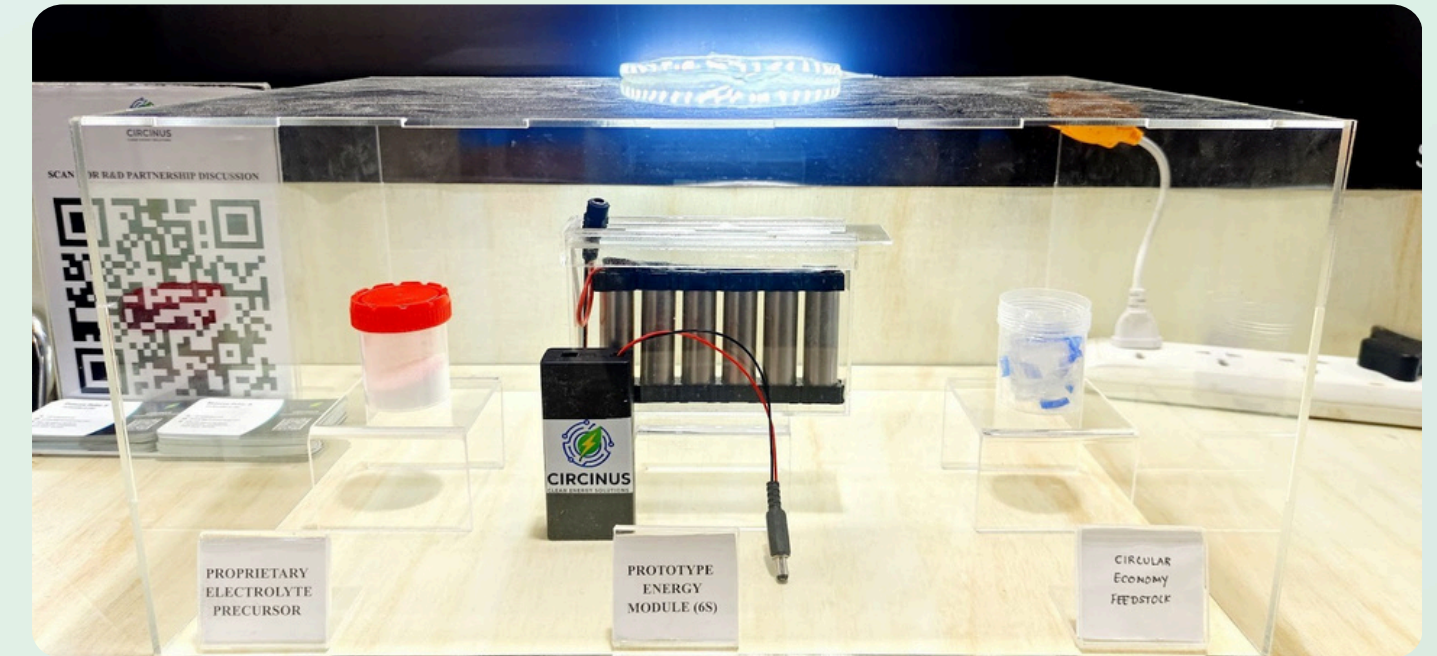
Mr. Ashwin Kumar P

Co-Founder & Chief of Operations

Oversees R&D operations, manufacturing protocols, and strategic partnerships across the Petrevolt ecosystem.



Our first alkaline battery prototype



Our updated prototype version-Rechargeable



Recognized by TN government & interviewed by top regional channel



Recognized by EDII-TN and grant received under "Innovation Voucher Program"

NATIONAL IMPACT & RISK ANALYSIS

STRATEGIC BENEFITS



Tactical Defence Edge

Rechargeable cells maintain 92% capacity at -40°C, operational down to -70°C — Cold-Start advantage for Siachen & Ladakh where standard batteries fail.



Circular Economy Leader

India formalizes the world's most successful large-scale plastic upcycling model — transforming 3.4M tonne liability into a strategic mineral reserve.



Energy Sovereignty

Indigenous technology ends technological vassalage, replacing billions in annual mineral imports with domestic waste-derived battery materials.

RISK / REWARD RATIO

⚠ THE RISK (MANAGEABLE)

Purely technical: scaling molecular harvesting of battery-grade materials from mixed waste. Government support mitigates this by funding industrial-grade SEM/XRD characterization to ensure material purity.

🚀 THE REWARD (EXPONENTIAL)

A multi-billion dollar domestic industry. Secures National IP and prevents the strategic trap of licensing our own technology back from abroad.

*"Rechargeable Sodium technology is not just an alternative;
it is India's sovereign energy solution."*

— Princy Polin S, Co-Founder & CEO, Petrevolt Energy Solutions

Let's build the future of Indian energy on domestic waste, for domestic needs.

PETREVOLT ENERGY SOLUTIONS PVT LTD

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